Questions to Add

AI Basics

1. **What is the difference between model training and inference?**
    a. **Training updates model parameters. inference uses a trained model to generate outputs**
    b. Training runs only on CPUs. inference runs only on GPUs
    c. Training stores datasets. inference cleans datasets
    d. Training and inference perform the same operation
2. **Which activity is an example of model inference?**

    A. Updating model weights using labelled examples
    B. Generating predictions using a trained model
    C. Calculating gradients during a training iteration
    D. Selecting hyperparameters using validation results
3. **A customer wants to deploy an AI model on a GPU instance. Which information is most useful when selecting the instance?**

    A. The number of users registered with the customer
    B. The programming language alone
    C. The size of the customer's source-code repository
    D. The model's memory requirements, workload type, and performance targets
4. **What does retrieval-augmented generation (RAG) typically do?**

    A. Updates model weights whenever a document is uploaded
    B. Replaces document retrieval with additional model training
    C. Supplies retrieved information as context for generation
    D. Converts generated answers into labelled training examples
5. **What distinguishes supervised learning from unsupervised learning?**

    A. Supervised learning requires GPUs; unsupervised learning requires CPUs
    B. Supervised learning uses labelled examples to learn a mapping
    C. Supervised learning uses less data for every type of problem
    D. Supervised learning produces rules instead of numerical predictions
6. **What distinguishes a hyperparameter from a learned parameter?**

    A. Hyperparameters are always calculated from test results
    B. Hyperparameters change automatically with every prediction
    C. Hyperparameters configure training rather than being learned as weights
    D. Hyperparameters exist only in models using neural networks

7. **Can an LLM run on a CPU?**

   A. Yes, but it must be trained again first

   B. Yes, with sufficient memory and compatible software

   C. No, it always requires a dedicated GPU

   D. No, it requires an external AI service

8. **Which evaluation best measures whether a RAG answer is grounded?**

   A. **Checking whether its claims are supported by the supplied sources**

   B. Checking whether its wording resembles previous model answers

   C. Checking whether it uses all available output tokens

   D. Checking whether it includes terminology from the user's question

9. **A team wants to use sensitive customer data with an AI service. What should the engineer do first?**

   A. Remove customer names and submit the remaining records

   B. Upload a small sample to check the model's response

   C. Verify data classification and the service's approved handling controls

   D. Encrypt the source file and share its password in the prompt

10. **What does an API contract typically define?**

    A. Request formats, response formats, and expected behaviours

    B. Server ownership, procurement costs, and renewal dates

    C. Model architecture, training examples, and learned weights

    D. User responsibilities, training schedules, and support hours

11. **Before building a Retrieval-Augmented Generation (RAG) system, what should the engineer do first?**

    A. Select a vector database and configure its search settings

    B. Define the use case and assess source quality and access permissions

    C. Choose an embedding model and generate document vectors

    D. Deploy an LLM endpoint and configure its response settings

12. **Why are application logs useful?**

    A. They improve the model's prediction accuracy

    B. They increase the server's processing capacity

    C. They prevent users from submitting invalid requests

    D. They help engineers investigate errors and events

13. **Where should an application's API keys and passwords be stored?**

    A. In the application's source code

    B. In a configuration file beside the source code

C. In a secure secrets manager

D. In container images within a private registry

14. **Which action should be avoided when using a public AI service?**

    A. Summarizing a company report published on its website

    B. Reviewing sample code containing fictional customer records

    C. Uploading internal customer records after removing names

    D. Drafting a proposal using publicly available information

15. **A team wants a custom GPT to answer questions from internal documents. What should be assessed first?**

    A. Whether all documents can fit within the upload limits

    B. Whether a larger model would produce better responses

    C. Whether intended users are authorized to access the documents

    D. Whether the instructions cover every possible user question

16. **Can every multimodal AI model process every type of input?**

    A. Yes, provided the input files are small

    B. No, supported input types depend on the model

    C. Yes, provided the server has enough memory

    D. No, each model accepts only one input type

17. **A client requests the largest available LLM for classifying support tickets. What should the engineer do?**

    A. Select it because parameter count determines task accuracy

    B. Choose a smaller model based only on the ticket length

    C. Use the existing model without further task evaluation

    D. Compare suitable models against quality, latency, and cost targets

18. **A client wants AI to summarize recorded meetings, including discussion and displayed slides. What should the engineer evaluate?**

    A. Whether a text model can infer slide details from meeting titles

    B. Whether image classification can replace speech transcription

    C. Whether processing every video frame improves all summary tasks

    D. Whether the pipeline captures and aligns relevant audio and visuals

19. **A RAG chatbot gives an outdated answer even though the source document was updated yesterday. What is the best initial investigation?**

    A. Trace the document through ingestion, indexing, retrieval, and caching

    B. Increase the number of retrieved passages for every question

    C. Fine-tune the model using the updated document

    D. Replace the embedding model with a larger alternative

20. **What is the main purpose of reviewing platform access logs?**
    a. **To identify unusual access activities**
    b. To measure application response time
    c. To estimate platform storage usage
    d. To monitor AI model performance
21. **An application consists of a frontend, load balancer, backend API, and AI service. Which request flow is most appropriate?**
    a. Frontend → AI Service → Load Balancer → Backend API
    b. **Frontend → Load Balancer → Backend API → AI Service**
    c. Frontend → Backend API → Load Balancer → AI Service
    d. Frontend → AI Service → Backend API → Load Balancer
22. **What is a token in an LLM?**
    a. **A Unit of text processed**
    b. An authentication password
    c. A GPU processing core
    d. A database record
23. **After rebuilding a Docker application, the old version still appears. Which command should be used to ensure the latest changes are included in a fresh image build?**
    a. docker compose build --pull
    b. **docker compose build --no-cache**
    c. docker compose build --force-rm
    d. docker compose build --progress plain
24. **Before storing extracted PDF text in a vector-based knowledge base, what should be performed?**
    a. Tokenization
    b. Fine-tuning
    c. **Text Embedding**
    d. Quantization
25. **You deployed an LLM on a VM server. What is the most appropriate way to allow other developers to integrate it into their applications?**
    a. Configure a reverse proxy to route model requests
    b. **Expose an authenticated inference API endpoint**
    c. Enable remote access to the model runtime
    d. Configure a load balancer for model traffic
26. **What does "B" represent in LLM models such as 7B, 14B, and 70B, and how does it affect performance?**

a. Billions of parameters; higher B always provides better accuracy and faster inference
b. Billions of parameters; lower B generally requires more GPU memory but offers faster inference
c. **Billions of parameters; higher B generally increases model capacity and computational requirements**
d. Billions of parameters; higher B primarily increases context window size and token generation speed

27. **A web application forwards user requests to an internal API. How should the backend verify the user's identity?**
    a. Validate the user ID provided in the request body
    b. Identify the user through the originating IP address
    c. **Validate the authenticated session or access token**
    d. Verify the username forwarded in a custom HTTP header
28. **A RAG system retrieves relevant text chunks but frequently loses important context from surrounding paragraphs. Which approach best addresses this?**
    a. Reduce chunk sizes without retaining relationships between chunks
    b. Increase embedding dimensions for all documents
    c. **Use parent-child chunking or retrieve neighboring chunks**
    d. Replace vector retrieval with document-level similarity scores
29. **Before generating embeddings from newly collected documents, what should a data pipeline prioritize?**
    a. Automatically replacing the existing vector index
    b. Increasing the embedding model's context window
    c. **Validating document content, metadata, and access permissions**
    d. Sending all documents directly to the language model
30. **A Docker container is running, but the application cannot be accessed externally. What should be investigated first?**
    a. Check container health and application logs
    b. **Verify port mappings and listening interfaces**
    c. Inspect Docker network and DNS settings
    d. Review container restart and resource policies
31. **A newly deployed container version starts successfully but causes application errors. The previous version was stable. What is the most appropriate recovery action?**
    a. Restart the container with updated settings
    b. Reconfigure the container's runtime environment
    c. **Restore the previous image and configuration**

d. Recreate the container with existing volumes

32. **An AI application is running, but GPU utilisation remains at 0% during inference. What should be verified first?**
    a. Whether the model's context length is configured
    b. **Whether the runtime detects and uses the GPU**
    c. Whether the API endpoint accepts requests
    d. Whether the model weights are quantized
33. **A model requires more GPU memory than the available accelerator provides. Which approach can reduce memory consumption?**
    a. Increase the maximum output tokens
    b. Increase the number of model replicas
    c. **Apply a lower-precision quantization format**
    d. Increase the inference batch size
34. **An AI model responds quickly for one user but becomes significantly slower under concurrent requests. What should be evaluated first?**
    a. Model checkpoint naming conventions
    b. **GPU utilisation and request scheduling**
    c. API documentation and response schemas
    d. Container restart and logging policies
35. **An AI platform service is running, but users report intermittent failures. What should be investigated first?**
    a. **Review service logs and error patterns**
    b. Increase the service timeout settings
    c. Restart the affected service instances
    d. Adjust the load balancer distribution
36. **An AI application reports connection timeouts when communicating with its database. What should be verified first?**
    a. Application interface configuration
    b. **Database connectivity and service availability**
    c. Model inference and GPU utilisation
    d. User session and authentication settings
37. **An engineer resolves a recurring application issue. What should be documented?**
    a. Only the date the issue occurred
    b. Only the final service status
    c. **The root cause and resolution steps**
    d. The engineer's troubleshooting duration

38. **Two AI inference services share one GPU. Both run successfully alone but encounter memory errors when started together. What is the most appropriate solution?**
    a. Increase the API connection timeout
    b. **Adjust GPU memory allocation or workloads**
    c. Increase the container's CPU limits
    d. Configure additional network interfaces
39. **A GPU has 16 GB VRAM. An engineer wants to deploy a 14B parameter LLM. Which factor is most important when assessing feasibility?**
    a. Model parameter count and API response format
    b. **Model quantization, memory overhead, and context length**
    c. Model training dataset size and tokenizer vocabulary
    d. Model benchmark ranking and supported languages
40. **An organization replaces its embedding model with another model that produces vectors of the same dimension and uses cosine similarity. However, retrieval quality drops significantly after deployment. What is the most likely cause?**
    a. The vector database requires additional index replicas
    b. The similarity threshold must match both tokenizers
    c. The embedding batch size affects retrieval accuracy
    d. **The embedding models use incompatible vector spaces**

Answer Sheet:

1-A,2-B,3-D,4-C,5-B,6-C,7-B,8-A,9-C,10-A,11-B,12-D,13-C,14-A,15-C,16-D,17-D,18-D,19-A,20-A,21-B,22-A,23-B,24-C,25-B,26-C,27-C,28-C,29-C,30-B,31-C,32-B,33-C,34-B,35-A,36-B,37-C,38-B,39-B,40-D